Newman-Penrose Formalism Calculations

Stephani et al., Equation (23.11)

Metric

$$ds^2 = -Ae^{(nt+nx)}\mathrm{d}t \otimes \mathrm{d}t + Be^{(nt+nx)}\mathrm{d}x \otimes \mathrm{d}x + e^{(\lambda t+nt-\lambda x+nx-t-x)}\mathrm{d}y \otimes \mathrm{d}y + e^{(-\lambda t+nt+\lambda x+nx)}\mathrm{d}z \otimes \mathrm{d}z$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Metric Functions

$$A = e^{\left(nt-nx+ae^{\left(-nt+nx+\frac{1}{2}\,t-\frac{1}{2}\,x\right)}+b\right)}$$

$$B = -\frac{(1+2\lambda)}{(1-2\lambda)}A$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}}\sqrt{A}e^{\left(\frac{1}{2}\,nt+\frac{1}{2}\,nx\right)}\right)\,dt + \left(-\sqrt{\frac{1}{2}}\sqrt{B}e^{\left(\frac{1}{2}\,nt+\frac{1}{2}\,nx\right)}\right)\,dx$$

$$n = \left(\sqrt{\frac{1}{2}}\sqrt{A}e^{\left(\frac{1}{2}\,nt+\frac{1}{2}\,nx\right)}\right)\,dt + \left(\sqrt{\frac{1}{2}}\sqrt{B}e^{\left(\frac{1}{2}\,nt+\frac{1}{2}\,nx\right)}\right)\,dx$$

$$m = \left(\sqrt{\frac{1}{2}}e^{\left(\frac{1}{2}\,\lambda t+\frac{1}{2}\,nt-\frac{1}{2}\,\lambda x+\frac{1}{2}\,nx-\frac{1}{2}\,t-\frac{1}{2}\,x\right)}\right)\,dy + \left(i\sqrt{\frac{1}{2}}e^{\left(-\frac{1}{2}\,\lambda t+\frac{1}{2}\,nt+\frac{1}{2}\,\lambda x+\frac{1}{2}\,nx\right)}\right)\,dz$$

$$\overline{m} = \left(\sqrt{\frac{1}{2}}e^{\left(\frac{1}{2}\,\lambda t+\frac{1}{2}\,nt-\frac{1}{2}\,\lambda x+\frac{1}{2}\,nx-\frac{1}{2}\,t-\frac{1}{2}\,x\right)}\right)\,dy + \left(-i\,\sqrt{\frac{1}{2}}e^{\left(-\frac{1}{2}\,\lambda t+\frac{1}{2}\,nt+\frac{1}{2}\,\lambda x+\frac{1}{2}\,nx\right)}\right)\,dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{2\,\sqrt{A}}\right)\,dt + \left(-\frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{2\,\sqrt{B}}\right)\,dx$$

$$n = \left(-\frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{2\,\sqrt{A}}\right)\,dt + \left(\frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{2\,\sqrt{B}}\right)\,dx$$

$$m = \left(\frac{1}{2}\,\sqrt{2}e^{\left(-\frac{1}{2}\,\lambda t-\frac{1}{2}\,nt+\frac{1}{2}\,\lambda x-\frac{1}{2}\,nx+\frac{1}{2}\,t+\frac{1}{2}\,x\right)}\right)\,dy + \left(\frac{1}{2}i\,\sqrt{2}e^{\left(\frac{1}{2}\,\lambda t-\frac{1}{2}\,nt-\frac{1}{2}\,\lambda x-\frac{1}{2}\,nx\right)}\right)\,dz$$

$$\overline{m} = \left(\frac{1}{2}\,\sqrt{2}e^{\left(-\frac{1}{2}\,\lambda t-\frac{1}{2}\,nt+\frac{1}{2}\,\lambda x-\frac{1}{2}\,nx+\frac{1}{2}\,t+\frac{1}{2}\,x\right)}\right)\,dy + \left(-\frac{1}{2}i\,\sqrt{2}e^{\left(\frac{1}{2}\,\lambda t-\frac{1}{2}\,nt-\frac{1}{2}\,\lambda x-\frac{1}{2}\,nx\right)}\right)\,dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{2\,\sqrt{A}}\,\partial_t - \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{2\,\sqrt{B}}\,\partial_x$$

$$\Delta = -\frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{2\,\sqrt{A}}\,\partial_t + \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{2\,\sqrt{B}}\,\partial_x$$

$$\delta = \frac{1}{2}\,\sqrt{2}e^{\left(-\frac{1}{2}\,\lambda t-\frac{1}{2}\,nt+\frac{1}{2}\,\lambda x-\frac{1}{2}\,nx+\frac{1}{2}\,t+\frac{1}{2}\,x\right)}\,\partial_y + \frac{1}{2}i\,\sqrt{2}e^{\left(\frac{1}{2}\,\lambda t-\frac{1}{2}\,nt-\frac{1}{2}\,\lambda x-\frac{1}{2}\,nx\right)}\,\partial_z$$

$$\overline{\delta} = \frac{1}{2}\,\sqrt{2}e^{\left(-\frac{1}{2}\,\lambda t-\frac{1}{2}\,nt+\frac{1}{2}\,\lambda x-\frac{1}{2}\,nx+\frac{1}{2}\,t+\frac{1}{2}\,x\right)}\,\partial_y - \frac{1}{2}i\,\sqrt{2}e^{\left(\frac{1}{2}\,\lambda t-\frac{1}{2}\,nt-\frac{1}{2}\,\lambda x-\frac{1}{2}\,nx\right)}\,\partial_z$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}ne^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{4\,\sqrt{A}} + \frac{\sqrt{2}ne^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{4\,\sqrt{B}} - \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{8\,\sqrt{A}} - \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{8\,\sqrt{B}}$$

$$\sigma = \frac{\sqrt{2}\lambda e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{4\,\sqrt{A}} - \frac{\sqrt{2}\lambda e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{4\,\sqrt{B}} - \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{8\,\sqrt{A}} - \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{8\,\sqrt{B}}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}ne^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{4\,\sqrt{A}} + \frac{\sqrt{2}ne^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{4\,\sqrt{B}} + \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{8\,\sqrt{A}} - \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{8\,\sqrt{B}}$$

$$\lambda = -\frac{\sqrt{2}\lambda e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{4\,\sqrt{A}} - \frac{\sqrt{2}\lambda e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{4\,\sqrt{B}} + \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{8\,\sqrt{A}} - \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{8\,\sqrt{B}}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}ne^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{8\,\sqrt{A}} - \frac{\sqrt{2}ne^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{8\,\sqrt{B}} - \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}\frac{\partial}{\partial x}A}{8\,A\sqrt{B}} - \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}\frac{\partial}{\partial t}B}{8\,\sqrt{A}B}$$

$$\gamma = \frac{\sqrt{2}ne^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{8\,\sqrt{A}} - \frac{\sqrt{2}ne^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}}{8\,\sqrt{B}} - \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}\frac{\partial}{\partial x}A}{8\,A\sqrt{B}} + \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,nt-\frac{1}{2}\,nx\right)}\frac{\partial}{\partial t}B}{8\,\sqrt{A}B}$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = &-\frac{\lambda^2 e^{(-nt-nx)}}{8\,A} + \frac{n^2 e^{(-nt-nx)}}{8\,A} - \frac{\lambda^2 e^{(-nt-nx)}}{8\,B} + \frac{n^2 e^{(-nt-nx)}}{8\,B} + \frac{\lambda^2 e^{(-nt-nx)}}{4\,\sqrt{A}\sqrt{B}} + \frac{n^2 e^{(-nt-nx)}}{4\,\sqrt{A}\sqrt{B}} + \frac{\lambda e^{(-nt-nx)}}{8\,A} - \frac{\lambda e^{(-nt-nx)}}{8\,B} \\ &+ \frac{ne^{(-nt-nx)}\frac{\partial}{\partial t}A}{8\,A^2} + \frac{ne^{(-nt-nx)}\frac{\partial}{\partial x}A}{8\,AB} + \frac{ne^{(-nt-nx)}\frac{\partial}{\partial x}A}{4\,A^{\frac{3}{2}}\sqrt{B}} + \frac{ne^{(-nt-nx)}\frac{\partial}{\partial t}B}{4\,\sqrt{A}B^{\frac{3}{2}}} + \frac{ne^{(-nt-nx)}\frac{\partial}{\partial t}B}{8\,AB} + \frac{ne^{(-nt-nx)}\frac{\partial}{\partial x}B}{8\,B^2} - \frac{e^{(-nt-nx)}}{16\,A} \\ &- \frac{e^{(-nt-nx)}}{16\,B} - \frac{e^{(-nt-nx)}}{8\,\sqrt{A}\sqrt{B}} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}A}{16\,A^2} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A}{16\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A}{8\,A^{\frac{3}{2}}\sqrt{B}} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B}{8\,\sqrt{A}B^{\frac{3}{2}}} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B}{16\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}B}{16\,B^2}\end{aligned}$$

$$\Phi_{01} = 0$$

$$\begin{aligned}\Phi_{02} = &\frac{\lambda ne^{(-nt-nx)}}{4\,A} + \frac{\lambda ne^{(-nt-nx)}}{4\,B} - \frac{\lambda e^{(-nt-nx)}}{8\,A} - \frac{ne^{(-nt-nx)}}{8\,A} - \frac{\lambda e^{(-nt-nx)}}{8\,B} + \frac{ne^{(-nt-nx)}}{8\,B} - \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial t}A}{8\,A^2} + \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial x}A}{8\,AB} \\ &+ \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial t}B}{8\,AB} - \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial x}B}{8\,B^2} + \frac{e^{(-nt-nx)}}{16\,A} - \frac{e^{(-nt-nx)}}{16\,B} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}A}{16\,A^2} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A}{16\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B}{16\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}B}{16\,B^2}\end{aligned}$$

$$\Phi_{10} = 0$$

$$\begin{aligned}\Phi_{11} = &-\frac{\lambda^2 e^{(-nt-nx)}}{16\,A} + \frac{n^2 e^{(-nt-nx)}}{16\,A} + \frac{\lambda^2 e^{(-nt-nx)}}{16\,B} - \frac{n^2 e^{(-nt-nx)}}{16\,B} + \frac{\lambda e^{(-nt-nx)}}{16\,A} - \frac{ne^{(-nt-nx)}}{16\,A} + \frac{\lambda e^{(-nt-nx)}}{16\,B} + \frac{ne^{(-nt-nx)}}{16\,B} + \frac{ne^{(-nt-nx)}\frac{\partial}{\partial t}A}{16\,A^2} + \frac{ne^{(-nt-nx)}\frac{\partial}{\partial x}A}{16\,AB} \\ &- \frac{ne^{(-nt-nx)}\frac{\partial}{\partial t}B}{16\,AB} - \frac{ne^{(-nt-nx)}\frac{\partial}{\partial x}B}{16\,B^2} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A^2}{16\,A^2B} + \frac{e^{(-nt-nx)}\frac{\partial^2}{(\partial x)^2}A}{8\,AB} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}A\frac{\partial}{\partial t}B}{16\,A^2B} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B^2}{16\,AB^2} - \frac{e^{(-nt-nx)}\frac{\partial^2}{(\partial t)^2}B}{8\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A\frac{\partial}{\partial x}B}{16\,AB^2}\end{aligned}$$

$$\Phi_{12} = 0$$

$$\begin{aligned}\Phi_{20} = &\frac{\lambda ne^{(-nt-nx)}}{4\,A} + \frac{\lambda ne^{(-nt-nx)}}{4\,B} - \frac{\lambda e^{(-nt-nx)}}{8\,A} - \frac{ne^{(-nt-nx)}}{8\,A} - \frac{\lambda e^{(-nt-nx)}}{8\,B} + \frac{ne^{(-nt-nx)}}{8\,B} - \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial t}A}{8\,A^2} + \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial x}A}{8\,AB} \\ &+ \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial t}B}{8\,AB} - \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial x}B}{8\,B^2} + \frac{e^{(-nt-nx)}}{16\,A} - \frac{e^{(-nt-nx)}}{16\,B} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}A}{16\,A^2} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A}{16\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B}{16\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}B}{16\,B^2}\end{aligned}$$

$$\Phi_{21} = 0$$

$$\begin{aligned}\Phi_{22} = &-\frac{\lambda^2 e^{(-nt-nx)}}{8\,A} + \frac{n^2 e^{(-nt-nx)}}{8\,A} - \frac{\lambda^2 e^{(-nt-nx)}}{8\,B} + \frac{n^2 e^{(-nt-nx)}}{8\,B} - \frac{\lambda^2 e^{(-nt-nx)}}{4\sqrt{A}\sqrt{B}} - \frac{n^2 e^{(-nt-nx)}}{4\sqrt{A}\sqrt{B}} + \frac{\lambda e^{(-nt-nx)}}{8\,A} - \frac{\lambda e^{(-nt-nx)}}{8\,B} \\ &+ \frac{n e^{(-nt-nx)}\frac{\partial}{\partial t}A}{8\,A^2} + \frac{n e^{(-nt-nx)}\frac{\partial}{\partial x}A}{8\,AB} - \frac{n e^{(-nt-nx)}\frac{\partial}{\partial x}A}{4\,A^{\frac{3}{2}}\sqrt{B}} - \frac{n e^{(-nt-nx)}\frac{\partial}{\partial t}B}{4\sqrt{A}B^{\frac{3}{2}}} + \frac{n e^{(-nt-nx)}\frac{\partial}{\partial t}B}{8\,AB} + \frac{n e^{(-nt-nx)}\frac{\partial}{\partial x}B}{8\,B^2} - \frac{e^{(-nt-nx)}}{16\,A} \\ &- \frac{e^{(-nt-nx)}}{16\,B} + \frac{e^{(-nt-nx)}}{8\sqrt{A}\sqrt{B}} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}A}{16\,A^2} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A}{16\,AB} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A}{8\,A^{\frac{3}{2}}\sqrt{B}} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B}{8\sqrt{A}B^{\frac{3}{2}}} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B}{16\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}B}{16\,B^2}\end{aligned}$$

$$\begin{aligned}\Lambda = &\frac{\lambda^2 e^{(-nt-nx)}}{48\,A} + \frac{n^2 e^{(-nt-nx)}}{16\,A} - \frac{\lambda^2 e^{(-nt-nx)}}{48\,B} - \frac{n^2 e^{(-nt-nx)}}{16\,B} - \frac{\lambda e^{(-nt-nx)}}{48\,A} - \frac{n e^{(-nt-nx)}}{16\,A} - \frac{\lambda e^{(-nt-nx)}}{48\,B} + \frac{n e^{(-nt-nx)}}{16\,B} - \frac{n e^{(-nt-nx)}\frac{\partial}{\partial t}A}{16\,A^2} \\ &- \frac{n e^{(-nt-nx)}\frac{\partial}{\partial x}A}{16\,AB} + \frac{n e^{(-nt-nx)}\frac{\partial}{\partial t}B}{16\,AB} + \frac{n e^{(-nt-nx)}\frac{\partial}{\partial x}B}{16\,B^2} + \frac{e^{(-nt-nx)}}{48\,A} - \frac{e^{(-nt-nx)}}{48\,B} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}A}{48\,A^2} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A}{48\,AB} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A^2}{48\,A^2B} \\ &- \frac{e^{(-nt-nx)}\frac{\partial^2}{(\partial x)^2}A}{24\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B}{48\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}A\frac{\partial}{\partial t}B}{48\,A^2B} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B^2}{48\,AB^2} + \frac{e^{(-nt-nx)}\frac{\partial^2}{(\partial t)^2}B}{24\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}B}{48\,B^2} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A\frac{\partial}{\partial x}B}{48\,AB^2}\end{aligned}$$

Weyl Tensor Components

$$\begin{aligned}\Psi_0 = &\frac{\lambda e^{(-nt-nx)}}{8\,A} - \frac{\lambda e^{(-nt-nx)}}{8\,B} + \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial t}A}{8\,A^2} - \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial x}A}{8\,AB} + \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial x}A}{4\,A^{\frac{3}{2}}\sqrt{B}} - \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial t}B}{4\sqrt{A}B^{\frac{3}{2}}} + \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial t}B}{8\,AB} - \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial x}B}{8\,B^2} \\ &- \frac{e^{(-nt-nx)}}{16\,A} - \frac{e^{(-nt-nx)}}{16\,B} - \frac{e^{(-nt-nx)}}{8\sqrt{A}\sqrt{B}} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}A}{16\,A^2} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A}{16\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A}{8\,A^{\frac{3}{2}}\sqrt{B}} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B}{8\sqrt{A}B^{\frac{3}{2}}} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B}{16\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}B}{16\,B^2}\end{aligned}$$

$$\Psi_1 = 0$$

$$\begin{aligned}\Psi_2 = &\frac{\lambda^2 e^{(-nt-nx)}}{12\,A} - \frac{\lambda^2 e^{(-nt-nx)}}{12\,B} - \frac{\lambda e^{(-nt-nx)}}{12\,A} - \frac{\lambda e^{(-nt-nx)}}{12\,B} + \frac{e^{(-nt-nx)}}{48\,A} - \frac{e^{(-nt-nx)}}{48\,B} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}A}{48\,A^2} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A}{48\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A^2}{24\,A^2B} \\ &+ \frac{e^{(-nt-nx)}\frac{\partial^2}{(\partial x)^2}A}{12\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B}{48\,AB} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}A\frac{\partial}{\partial t}B}{24\,A^2B} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B^2}{24\,AB^2} - \frac{e^{(-nt-nx)}\frac{\partial^2}{(\partial t)^2}B}{12\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}B}{48\,B^2} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A\frac{\partial}{\partial x}B}{24\,AB^2}\end{aligned}$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{\lambda e^{(-nt-nx)}}{8\,A} - \frac{\lambda e^{(-nt-nx)}}{8\,B} + \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial t}A}{8\,A^2} - \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial x}A}{8\,AB} - \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial x}A}{4\,A^{\frac{3}{2}}\sqrt{B}} + \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial t}B}{4\,\sqrt{A}B^{\frac{3}{2}}} + \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial t}B}{8\,AB} - \frac{\lambda e^{(-nt-nx)}\frac{\partial}{\partial x}B}{8\,B^2}$$
$$- \frac{e^{(-nt-nx)}}{16\,A} - \frac{e^{(-nt-nx)}}{16\,B} + \frac{e^{(-nt-nx)}}{8\,\sqrt{A}\sqrt{B}} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}A}{16\,A^2} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A}{16\,AB} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}A}{8\,A^{\frac{3}{2}}\sqrt{B}} + \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B}{8\,\sqrt{A}B^{\frac{3}{2}}} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial t}B}{16\,AB} - \frac{e^{(-nt-nx)}\frac{\partial}{\partial x}B}{16\,B^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"